Intelligent Complaint Classification Using Localized Transformer Architectures

Project Interest

Intelligent customer complaint routing in the e-commerce sector.

Research Topic

Intelligent Complaint Classification Using Localized Transformer Architectures.

Community

Online consumer and e-commerce community.

Target Sector

Business Administration (E-commerce).

Target Users

E-commerce customers and customer support personnel.

Values Framework

Accessibility, efficiency, fairness, customer satisfaction, transparency, accountability, and responsible AI.

Community Problem

E-commerce businesses receive large volumes of customer complaints through online channels, including websites, email, live chat, and mobile applications. Complaints may involve payments, orders, deliveries, refunds, returns, accounts, or products. Many support systems still depend on manual categorization, predefined options, or keyword-based routing. These approaches may fail to understand the actual meaning of a customer's complaint, especially when messages contain spelling errors, informal language, or multiple issues. As a result, complaints may be routed to the wrong category or support team, leading to longer response times, increased workload, and reduced customer satisfaction.

Evaluation

The identified problem aligns with the project's values by improving accessibility through natural-language complaint submission, increasing operational efficiency by reducing manual triage, promoting fair and consistent complaint classification, and improving the customer support experience. Transparency and accountability will be supported by clearly evaluating model performance and retaining human oversight for uncertain or sensitive complaints. The use of publicly available consumer complaint data also makes the problem suitable for developing and evaluating a Transformer-based classification model without requiring access to private company data.

Research Problem Statement

E-commerce businesses receive customer complaints through various digital support channels, including websites, emails, live chat, and mobile applications. Despite these multiple access points, many complaints are still categorized through manual review, predefined options, or simple keyword-based processes that may not accurately capture the customer's intended issue. Complaints involving payments, orders, deliveries, refunds, returns, accounts, and products may therefore be incorrectly categorized or routed, increasing response times and placing additional workload on customer support personnel.

This research proposes the development of an intelligent complaint classification system using a localized transformer architecture, such as BERT or a distilled variant, to automatically classify customer complaints based on their textual content. The model will be trained and evaluated using a diverse, publicly available dataset of real-world consumer complaints. The scope of this research is limited to complaint classification and routing rather than complaint resolution.

The proposed system is expected to reduce manual triage, improve classification consistency, and support more efficient customer service operations. By applying transformer-based language models to a practical customer support challenge, this research aims to improve the efficiency and quality of complaint handling while maintaining human oversight for uncertain or sensitive cases.

Values We Are Committing To

Fairness: The model will be evaluated across common and less-represented complaint categories to identify the effects of class imbalance and reduce inconsistent performance.

Transparency: Model performance will be reported using clear evaluation metrics, including precision, recall, and F1-score, and limitations of the dataset and model will be documented.

Accountability: The system will support human decision-making rather than replace human judgement, particularly for uncertain, sensitive, or high-impact complaints.

Meaningful Impact: The project will focus on a practical customer-support problem where improved classification can reduce manual effort and support faster, more consistent complaint routing.